

Online Complaint Registration and Management System

Phase 5: Project Development Phase

SmartBridge Internship Project Submission

Team: Online Complaint Registration and Management System

5.1 Sprint-Wise Delivery

Sprint	Features Delivered
Sprint 1	Express server setup, MongoDB Atlas connection (Mongoose), User schema with bcrypt password hashing, /api/users/register and /api/users/login endpoints, JWT token generation, protect and authorize auth middleware.
Sprint 2	Complaint schema (title, description, category, location, priority, status, assignedAgent), Notification schema, full complaint CRUD (create, list with filters, get by ID, update, delete), automatic notification creation on complaint events, dashboard-stats aggregation endpoint.
Sprint 3	React app routing (App.jsx), Home/Login/Register pages, Dashboard page (complaint list with search & filters), Add Complaint form, Complaint Detail page with role-based status/agent-assignment controls, Axios API layer with JWT interceptor and 401 auto-logout.
Sprint 4	Admin Dashboard (analytics tab, complaints tab, user-management tab with role changes and deletion), responsive Navbar/Sidebar, deployment configuration for Render (backend Web Service + frontend Static Site), end-to-end testing and bug fixes.

5.2 API Endpoints Implemented

Method	Endpoint	Access	Description
POST	/api/users/register	Public	Register a new user
POST	/api/users/login	Public	Authenticate and receive a JWT
GET	/api/users/profile	Authenticated	Get the logged-in user's profile
GET	/api/users	Admin	List all users
GET	/api/users/agents	Admin	List all agents
PUT	/api/users/:id/role	Admin	Change a user's role
DELETE	/api/users/:id	Admin	Delete a user
POST	/api/complaints	Authenticated	Create a new complaint
GET	/api/complaints	Authenticated	List complaints (filtered by role)
GET	/api/complaints/:id	Authenticated	Get a single complaint
PUT	/api/complaints/:id	Owner / Agent / Admin	Update a complaint (details, status, assignment)
DELETE	/api/complaints/:id	Admin	Delete a complaint
GET	/api/complaints/stats	Admin	Dashboard analytics
GET	/api/complaints/notifications	Authenticated	List the user's notifications
PUT	/api/complaints/notifications/:id/read	Authenticated	Mark a notification as read

5.3 Security Measures Implemented

- Passwords hashed with bcryptjs before being stored (never stored in plaintext).
- JWT tokens signed with a server-side secret, expiring after 30 days.
- Role-based route protection (protect + authorize middleware) so only authorized roles can access sensitive endpoints (e.g. user management, analytics).
- Ownership checks on complaint access/update so a user cannot view or modify another user's complaint.
- Environment variables (.env) used for all secrets and connection strings, excluded from version control via .gitignore.

5.4 Screens Delivered

The following screens were implemented and should be captured as screenshots (from a locally running instance) for the final documentation and demonstration:

- Home page (landing page with call-to-action)
- Register and Login pages
- Dashboard — complaint list with search/filter
- Add Complaint — complaint submission form
- Complaint Detail — status and agent management
- Admin Dashboard — analytics overview, complaint management, user management

5.5 Source Code

The complete source code is maintained in the project GitHub repository, including backend (Node.js/Express/MongoDB) and frontend (React/Vite/Tailwind CSS) folders, along with setup instructions in README.md.

Repository: <https://github.com/parimivamsi543-png/online-complaint-system>